[INCH-POUND]
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FEDERAL SPECIFICATION

HAMMERS, HAND AND MALLETS, SURFACE PROTECTIVE; HOLDERS AND ADAPTERS, INSERTED HAMMER FACE, AND FACES, HAMMER, INSERTED

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hammers, mallets, adapters, and inserted faces which are used to strike and provide protection to the surfaces of metals and finished products.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, styles and designs. Hammers, mallets, adapters, and inserts shall be of the following types, classes, styles, and designs as specified (see 6.2):

Type I - Hammer, inserted face
- Class 1 - Split head
  - Style A - Copper inserts
  - Style B - Rawhide inserts
  - Style C - Lead inserts
  - Style D - Plastic inserts
  - Style E - Rubber inserts
- Class 2 - Press-on head
  - Style D - Plastic inserts
    - Design a - Regular
    - Design b - No bounce
- Class 3 - Screw-in head
  - Style D - Plastic inserts
    - Design a - Regular
    - Design b - No bounce

Type II - Hammer, solid head
- Class 1 - Copper
- Class 2 - Lead

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5120

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Type III - Mallet
Class 1 - Rawhide
Class 3 - Plastic
Class 4 - Rubber

Type IV - Adapter, Screw-in Head

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

NN-H-93	-	Handles; Hickory, Striking Tool
PPP-P-40	-	Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery
TT-S-735	-	Standard Test Fluids; Hydrocarbon

Federal Standards:

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-102	-	Preservation, Packaging and Packing Levels
FED-STD-601	-	Rubber, Sampling, and Testing

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft Missile, and Ordnance

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Consolidated Freight Classification Committee:

Consolidated Freight Classification Rules and Container Specifications

(Application for copies should be addressed to Consolidated Classification Committee, 202 Chicago Union Station, Chicago, IL 60606.)

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of hammers, mallets, adapters, and inserted faces otherwise conforming to this specification.

3.2 Material. The material shall be as hereinafter specified for the individual types. The chemical composition and the tempering or curing techniques employed shall be such as to produce tools that conform to this specification.

3.3 Design. The hammers, mallets, and adapters specified herein shall be of the latest design, convenient to use, and entirely adequate for the purpose intended (see 1.1). Under visual examination, the hammer shall be well proportioned, i.e., the two halves of the hammer shall correspond in size and shape. The longitudinal axis of the head shall be perpendicular to the axis of the handle, and the faces of the hammer shall be in proper alignment with the elliptical or other elongated cross sections of the handle and eye when applicable.

3.4 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.5 Finish. Hammers and mallets except handles shall have either a ground, machined, natural, or enamel finish. The adapters shall have a ground or machined finish.

3.5.1 Ground or machined finish. The ground or machined finish shall be a bright and evenly finished surface, free from deep scratches or excessive roughness.

3.5.2 Natural finish. The natural finish shall be an even, smooth cast, molded, extruded, rolled, or forged surface, free from scale, rust, other harmful imperfections.

3.5.3 Enameled finish. The enameled finish shall be of a waterproof paint or enamel applied to a surface. The coating shall not be tacky nor flake off readily.

3.6 Weight. Weight of type I hammers shall include holder (head and handle) and inserts. The weight of type II hammers and type III mallets shall include head and handle.

3.7 Rubber for inserts. Rubber for rubber inserts for hammers shall conform to Table I.

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TABLE I. Physical properties of rubber for rubber inserts

Tensile strength, psi	2500 (minimum)
Ultimate elongation, percent	2000 (minimum)
Shore A hardness, instantaneous	95 ± 5
Bashore resilience, 21st impact, percent	15 (maximum)
Graves tear strength, pounds per inch thickness	200 (minimum)
Cut resistance, pounds per inch, width	200 (minimum)
Impact-indentation, inch:	
At 82°F	0.04 (maximum)
At 120°F	0.04 (maximum)
Volume swell after 48 hours in ASTM Oil No. 3:	
At 158°F percent	10 (maximum)

3.8 Rubber for mallet heads. Rubber for mallet heads shall conform to Table II.

TABLE II. Physical properties of rubber for mallet heads

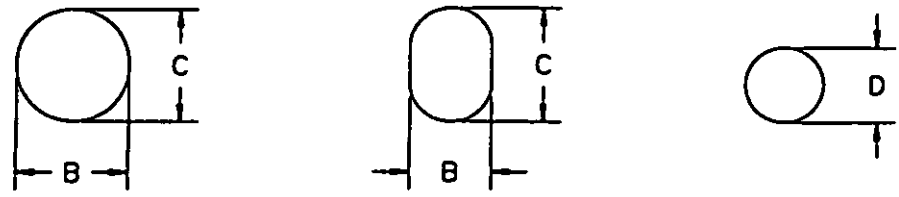
Tensile strength, psi	2500 (minimum)
Ultimate elongation, percent	10 (minimum)
Shore A hardness, instantaneous	95 ± 5
Bashore resilience, 21st impact, percent	28 (maximum)
Volume swell after 48 hours in ASTM Oil No. 3:	12 (maximum)
At 158°F percent	
Tear strength, pounds per inch, thickness	200 (minimum)

3.9 Eyes.

3.9.1 Tools with wood handles. The head (except for plastic and rubber mallets) shall be provided with an eye which shall extend through the head and in which the handle shall be fitted. Eyes shall have an inward taper at an angle of approximately 3° to permit proper wedging.

3.9.2 Eye numbers. The eye number specified for individual types shall be in substantial agreement with the respective eye number shown in Figure 1. (Eye numbers do not apply to some tools with steel or fiberglass handles.)

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EYE NO. 2 EYE NO. 3 EYE NO. 4

Figure 1. Eye numbers.

3.9.3 Tools with fiberglass-reinforced plastic handles. Hammer heads with fiberglass-reinforced plastic handles shall have either an eye as specified in 3.9.1 or a hole shall be drilled or reamed approximately three-quarters of the distance though the head in the place where the eye is normally located.

3.10 Handles. Each hammer or mallet shall be fitted with a well-proportioned wood or fiberglass handle, as specified (see 6.2), or steel handles as specified in 3.10.2.

3.10.1 Wood handles. Each hammer or mallet shall be fitted with a well-proportioned wood handle, unless hammers with steel handles are specified. The wood handle shall be made of hickory wood and shall be in accordance with NN-H-93, grade A, except Table I, column titled "Maximum admissible blemishes and defects", delete text except for "Four medium streaks or equivalent in shorter streaks. Light strain."

3.10.2 Steel handles. Type II, class 2 lead hammers shall be furnished with steel handles only. Type II, class 1 copper hammers shall be furnished with steel handles when specified in the contract or order. All steel handles shall be of forged steel ,or seamless, or welded steel tubing, well-proportioned and comfortable to the hand. The grip shall be deeply marked or knurled for safe usage. The head shall be molded securely around the handle, pinned, or otherwise secured in such a manner that the handle will not bend or loosen under any working condition. There shall be no openings between the handle and the head where the eye for the wood-handled tool is normally located. The grip portion of the handle may be covered with rubber, neoprene, or other comparable material.

3.10.3 Fiberglass-reinforced plastic handles. Fiberglass-reinforced plastic handles shall be well proportioned and shall be molded of a thermosetting type plastic containing a minimum of 60 percent fiberglass. Glass filament fiber orientation shall be unidirectional and longitudinal, and fibers shall be continuous. Fiberglass-reinforced plastic handles shall be inserted and bonded by means of a chemical adhesive, or otherwise affixed in eyes of tool heads in such a manner that the handle will not loosen under any working condition. There shall be no openings between the handle and head where the eye for the wood handles is normally located. The grip portion shall be covered with rubber, neoprene, or other comparable material. The eye section of the handle may be secured or bonded to the head by means of a chemical adhesive.

3.10.4 Replacement fiberglass-reinforced plastic handle kits. Replacement kits shall include the following : one (1) fiberglass-reinforced handle as specified in 3.10.3, of the type, class, weight, length, and eye size specified (see 6.2.1), one (1) ferrule cap, one ferrule collar, sufficient quantity of chemical adhesive bonding cement or other suitable bonding material, one (1) mixing container and stirring stick for bonding material, and one (1) set of instructions.

3.10.5 Chemical adhesive bonding. Handles secured by chemical adhesive bonding or means other than edging shall be capable of withstanding a temperature of - 30°F. When specified, (see 6.2 or 6.2.1, as applicable), chemical adhesive bonding or means other than wedging shall be capable of withstanding a temperature of - 60°F.

3.11 Assembly of wood handles to the head. Except as specified for the individual types, the eye section of the wood handle shall be secured with one or more metal wedges or metal screws, without or in combination with wood wedges. Wedges and screws shall be driven or screwed, respectively, flush with the front surface of the tool eye. The wedge, wedges, or screws, or a combination of the same, shall create sufficient expansion to completely fill the tool eye and tightly wedge the handle in the eye of the tool, so as to prevent the head from working loose under the

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most severe working conditions. The metal wedges shall be of the ribbed or corrugated type. The ribs or corrugation shall run in a horizontal or diagonal direction. The vertical direction is represented by the direction in which the wedge is to be driven. The small end of the wedge shall have a slight flatness to prevent splitting action of the tool handle. Screws shall be of the tapered, headless type. At the option of the manufacturer, handles may be secured and bonded to the head by means of a chemical adhesive that will prevent the handle from loosening under any normal working condition. Handles assembled with chemical adhesives shall be subjected to the pull-apart test (see 4.4.2.12).

3.12 Type I, hammer, inserted face. Type I hammers may be procured completely assembled or individual inserts and holders may be procured as replacement items (see 6.2).

3.12.1 Class 1, split head. Class 1 hammers shall consist essentially of two inserts of copper, rawhide, lead, plastic, or rubber, as specified (see 6.2), and a holder (head and handle). The inserts shall be cylindrical in shape with the striking surface of the face being either flat or slightly convex.

3.12.1.1 Style A, copper inserts. The inserts shall have a copper content of not less than 99 percent and conform to the applicable diameter of faces specified herein.

3.12.1.1.1 Holder (head and handle). The head shall be of malleable iron or steel of two-piece construction held together by a threaded connector and a tightening nut or by two countersunk cap screws. The head shall be fitted with a hickory handle conforming to 3.10.1 or a fiberglass handle conforming to 3.10.3. The handles shall be secured with wedges conforming to 3.11 except the wedges may be of wood. A steel pin shall be inserted transversely through the handle and the fixed section of the head and secured by riveting. In addition, a steel pin or screw shall be inserted transversely through the handle and the fixed section of the head secured by peening or tightening. The pin or screw shall be located adjacent to the threaded portion or eye side of the fixed section.

3.12.1.1.2 Type I, class 1, style A hammers. Type I, class 1, style A hammers shall conform to Table III for the weight specified and shall be similar to Figure 2 (see 6.2).

TABLE III. Type I, class 1, style A hammers, inserted face, split head, copper inserts

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
2	1-1/4	3-3/8	11	2, 3, or 4
2-3/4	1-1/2	3-5/8	12	2, 3, or 4
4	1-3/4	4	12	2, 3, or 4
4 (+ 10, - 6 oz.)	2	5-5/8 (+ 1, - 1/2)	13	2, 3, or 4
6-1/2	2	5-1/8	13	2, 3, or 4

1/ Weight includes head, inserts, and handle.

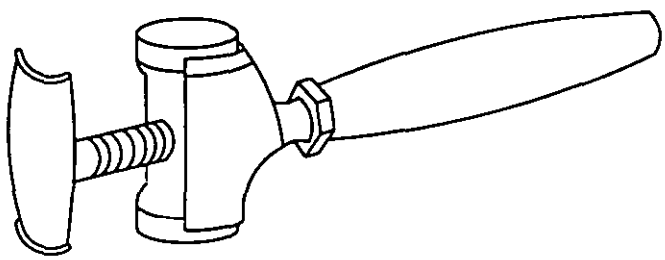


Figure 2. Type I, class 1, split head inserted face hammer.

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3.12.1.2 Style B, rawhide inserts. Inserts shall be made of one piece of rawhide, with not more than two filler strips. The hide shall be wound in a smooth, spiral, cylindrical shape. Joints at the end of the spiral shall be smooth and may be formed by a skived and glued lap, or a molded and pinned lap. If a pinned lap is used, it shall be not more than 3/16 inch off the center of the insert. The striking face shall show no voids and shall be parallel within 1/32 inch. It shall be not more than 2° off the perpendicular, measured at any position on the side of the insert. The edge of the face shall be radiused or chamfered. The sides shall be smooth and free from air pockets, except that cracks or wrinkles not over 1/16 inch deep and at least 1/4 inch from the striking face will be permitted. The inserts shall be completely covered with a moisture-resistant coating. The inserts shall conform to the applicable diameter of insert faces specified herein.

3.12.1.2.1 Holder (head and handle). The holder shall conform to 3.12.1.1.1.

3.12.1.2.2 Type I, class 1, style B hammers. Type I, class 1, style B hammers shall be similar to Figure 2 and shall conform to Table IV for the weight specified (see 6.2).

TABLE IV. Type I, class 1, style B hammers, inserted face, split head, rawhide inserts

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
1-1/4	1-1/4	3-3/8	11	2, 3, or 4
1-3/4	1-1/2	3-5/8	12	2, 3, or 4
2-3/4	1-3/4	4	12	2, 3, or 4
4-1/4	2	5-1/8	13	2, 3, or 4
6	2-3/4	5-7/8	14	2, 3, or 4

1/ Weight includes head, inserts, and handle.

3.12.1.3 Style C, lead inserts. The inserts shall be of antimonial lead having a Brinell hardness between 10 and 16 and shall be suitable for hammering purposes. The inserts shall conform to the applicable diameter of insert faces specified herein.

3.12.1.3.1 Holder (head and handle). The holder shall conform to 3.12.1.1.1.

3.12.1.3.2 Type I, class 1, style C hammers. Type I, class 1, style C hammers shall conform to Table V for the weight specified and shall be similar to Figure 2 (see 6.2).

TABLE V. Type I, class 1, style C hammers, inserted face, split head, lead inserts

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
2	1-1/4	3-3/8	11	2, 3, or 4
3	1-1/2	3-5/8	12	2, 3, or 4
4-1/2	1-3/4	4-1/8	12	2, 3, or 4
7	2	5-1/8	13	2, 3, or 4

1/ Weight includes head, inserts, and handle.

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3.12.1.4 Style D, plastic inserts. The inserts shall be of plastic or similar material and shall be molded around a metal or hickory disc to prevent the necessary rigidity to securely hold the insert under maximum working conditions. The inserts shall have a durometer hardness range between 73 Shore A, 30 second, and 76 Shore D, 30 second. The material shall withstand all test requirements of 4.4.2 except the impact test at - 30° (see 4.4.2.5) and the permanent distortion test (see 4.4.2.9.).

3.12.1.4.1 Holder (head and handle). The holder shall conform to 3.12.1.1.1.

3.12.1.4.2 Type I, class 1, style D hammers. Type I, class 1 style D hammers shall conform to Table IV for the weight specified and shall be similar to Figure 2 (see 6.2).

TABLE VI. Type I, class 1, style D hammers, inserted face, split head, plastic inserts

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
1-1/4	1-1/4	3-3/8	11	2, 3, or 4
1-5/8	1-1/2	3-5/8	12	2, 3, or 4
2-3/4	1-3/4	4	12	2, 3, or 4
4-1/4	2	5-1/8	13	2, 3, or 4

1/ Weight includes head, inserts, and handle.

3.12.1.5 Style E, rubber inserts. The rubber insert shall have a metal disc or washer vulcanized to one end to provide the necessary rigidity to securely hold the insert under maximum working conditions. The rubber shall conform to the physical requirements of 3.7 and the test requirements under 4.4.1.

3.12.1.5.1 Holder (head and handle). The holder shall conform to 3.12.1.1.1.

3.12.1.5.2 Type I, class 1, style E hammers. Type I, class 1, style E hammers shall be similar to Figure 2 and shall conform to Table VII for the weight specified (see 6.2).

TABLE VII. Type I, class 1, style E hammers, inserted face, split head, rubber inserts

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
1-3/8	1-1/4	3-3/8	11	2, 3, or 4
1-3/4	1-1/2	3-5/8	12	2, 3, or 4
3	1-3/4	4	12	2, 3, or 4
4-1/2	2	5-1/8	13	2, 3, or 4
6-1/2	2-3/4	5-1/4	14	2, 3, or 4

1/ Weight includes head, inserts, and handle.

3.12.2 Class 2, press-on head. Class 2 hammers shall consist essentially of two inserts of plastic and a holder (head and handle). The inserts shall be cylindrical in shape with the striking surface of the face being either flat or slightly convex.

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3.12.2.1 Style D plastic inserts. The inserts shall be of plastic or similar material and have a durometer hardness range between 85 Shore A, 30 second, and 76 Shore D, 30 second. The material shall withstand all test requirements of 4.4.2, except impact at - 30° (see 4.4.2.5) and permanent distortion test (see 4.4.2.9).

3.12.2.1.1 Holder (head and handle). The head shall be of aluminum alloy, zinc alloy, malleable iron, or steel. The design shall be such that it will hold the insets securely under all working conditions but will permit ready removal and replacement of damaged inserts. The head shall be of one- or two-piece construction. This type is suitable for yielding materials only, but the material shall not be softer than 85 on the Shore A scale. Inserts are secured by expanding on a central core while containing the expansion by a retaining ring. The expanding core shall be not less than half the diameter of the head nor extend into the inserts less than the width of the expanding ring. If the head is of two-piece construction, the retaining ring may be separate from the head, but shall be of the material having a tensile strength of not less than 15,000 pounds per square inch. If the head is of one-piece construction, the expanding core shall be contained within a shell, the end of which serves as a retaining ring. The head shall be fitted with a hickory handle conforming to 3.10.1 or fiberglass handle conforming to 3.9.3. The handles shall be secured with wedges conforming to 3.11 except the wedges may be of wood.

3.12.2.1.2 Design a, regular. Design a hammers shall conform to Table VIII for the weight specified and shall be similar to Figure 3 (see 6.2).

TABLE VIII. Type I, class 2, style D, design a hammers, inserted faces, press-on head, plastic insert faces, regular

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
5/8	1-1/4	3-3/4	11-1/2	2 or 3
1	1-1/2	4-1/4	12	2 or 3
1-1/4	1-1/2	4-1/2	12	2 or 3
1-1/2	1-3/4	5-1/2	13-1/4	2 or 3
1-3/4	2-1/4	5-1/2	14-3/4	2 or 3
2-1/2	2-1/4	6	14-3/4	2 or 3
3-1/4	2-1/4	6	14-3/4	2 or 3
3-3/4	2-1/4	6	14-3/4	2 or 3

1/ Weight includes head, inserts, and handle.

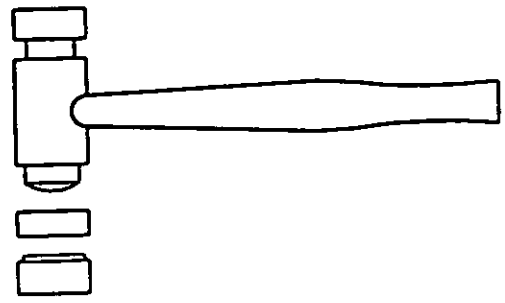


Figure 3. Type I, class 2, style D, design a and b hammers, inserted faces, press-on head, plastic insert faces, regular.

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3.12.2.1.3 Design b, no bounce. The style D, design b hammers shall conform to 3.12.2 through 3.12.2.1.1 except the head shall be designed to hold small metal pellets to restrict rebound during normal use but is not required to retain the pellets within the head when the handle is removed. Design b hammers shall conform to Table IX for the weight specified and shall be similar to Figure 3 (see 6.2).

TABLE IX. Type I, class 2, style D, design b hammers, inserted faces, press-on head, plastic insert faces, no bounce

Weight 1/ + 15 percent - 5 percent (Pounds)	Diameter of insert faces ± 1/16 (Inches)	Length of head including inserts + 3/4 - 1/4 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
2/3	1	3-3/4	11	2 or 3
5/8	1-1/4	3-3/4	11-1/2	2 or 3
1	1-1/2	4-1/4	12	2 or 3
1-1/2	1-3/4	5-1/2	13-1/4	2 or 3
2	2-1/4	5-1/2	13-3/4	2 or 3
4	2-1/4	6	13-1/4	2 or 3

1/ Weight includes head, inserts, and handle.

3.12.3 Class 3, screw-in head. Class 3 hammers shall consist essentially of two inserts of plastic and a holder (head and handle). The threads shall be Coarse Thread Series, Class 2B Unified or Class 2 American National in accordance with FED-STD-H28 and supplement thereto. The dimensions and tolerances shall be such that all inserts and heads of the same nominal diameter can be readily removed, replaced, and interchanged without requiring modification.

3.12.3.1 Style D, plastic inserts. The insert shall be of plastic or similar material and have a threaded stud molded or otherwise permanently fastened into the base of the insert for attachment to the head. Steel studs shall be protected against corrosion by zinc plating. When nonmagnetic studs are specified (see 6.2), the studs shall be of bronze or brass.

3.12.3.1.1 Hardness of plastic inserts. Plastic inserts shall be of seven individual hardness grades with each grade having an identifying color. Plastic inserts shall conform to Table X for the hardness name specified (see 6.2).

TABLE X. Type I, class 3, style D plastic insert hardness grades

Hardness name	Hardness color	Test instrument and reading	Hardness range
Super soft	Gray	Shore A,Instantaneous	45 to 58
Soft	Brown	Shore A, 30 second	45 to 58
Medium	Red	Shore A, 30 second	59 to 72
Tough	Green	Shore A, 30 second	73 to 85
Medium hard 1/	Cream	Shore D, 30 second	40 to 48
Hard 2/	Black	Shore D, 30 second	49 to 65
Extra hard 2/	Yellow	Shore D, 30 second	66 to 85

1/ Tests of 4.4.2.5, impact at -30°, and 4.4.2.9, permanent distortion, not applicable.
2/ Test of 4.4.2.9, permanent distortion, not applicable.

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3.12.3.1.2 Shapes and dimensions of plastic inerts. Plastic inserts shall be of five individual face shapes and diameters. Unless otherwise specified in the contract or order, the standard face shape shall be furnished. The outside diameter of the standard face shape may be straight or slightly tapered. The insert diameter shall be as specified (see 6.2). The dimensions on the inserts, identified by the letters A through H, shall conform to Table XI for the face shape and inserted diameter specified, and be similar to Figures 4A through 4E (see 6.2).

TABLE XI. Dimensions of type I, class 3, style D plastic inserts

Figure number and face shape	A Diameter ± 1/16 (Inches)	B Seat diameter 1/ (Inches)	C Seat depth +.030 -.000 (Inch)	D Length of stud ± 1/16 (Inch)	E Diameter of threads nominal (Inch)	F Useable face ± 1/16 (Inches)	G Radius ± 1/65 (Inch)	H Angle ± 1 (Degrees)
4A Standard	1	0.875	0.187	3/8	1/4	0.615	1/32	-
4B Dome	1	0.875	0.187	3/8	1/4	0.615	3/16	-
4C Wedge	1	0.875	0.187	3/8	1/4	0.735	1/8	76
4D Square	1	0.875	0.187	3/8	1/4	0.735	3/32	-
4E Conical	1	0.875	0.187	3/8	1/4	0.840	1/16	23
4A Standard	1-1/2	1.250	0.250	3/8	5/16	0.650	1/32	-
4B Dome	1-1/2	1.250	0.250	3/8	5/16	0.650	1/4	-
4C Wedge	1-1/2	1.250	0.250	3/8	5/16	1.225	1/2	76
4D Square	1-1/2	1.250	0.250	3/8	5/16	1.225	7/64	-
4E Conical	1-1/2	1.250	0.250	3/8	5/16	1.275	1/16	23
4A Standard	2	1.750	0.312	3/8	3/8	0.970	1/16	-
4B Dome	2	1.750	0.312	3/8	3/8	0.970	3/8	-
4C Wedge	2	1.750	0.312	3/8	3/8	1.500	3/16	76
4D Square	2	1.750	0.312	3/8	3/8	1.500	3/32	-
4E Conical	2	1.750	0.312	3/8	3/8	1.500	3/32	34
4A Standard	2-1/2	2.000	0.375	9/16	7/16	1.125	3/32	-
4B Dome	2-1/2	2.000	0.375	9/16	7/16	1.125	7/16	-
4C Wedge	2-1/2	2.000	0.375	9/16	7/16	1.750	3/16	76
4D Square	2-1/2	2.000	0.375	9/16	7/16	1.750	3/16	-
4E Conical	2-1/2	2.000	0.375	9/16	7/16	1.750	3/32	34
4A Standard	3	2.500	0.500	9/16	1/2	1.250	3/16	-
4B Dome	3	2.500	0.500	9/16	1/2	1.250	1/2	-
4C Wedge	3	2.500	0.500	9/16	1/2	2.000	1/4	76
4D Square	3	2.500	0.500	9/16	1/2	2.000	7/32	-
4E Conical	3	2.500	0.500	9/16	1/2	2.000	1/8	34

1/ Tolerance for seat diameters shall be as follows:
+ 0.000, - 0.015 for tough and softer grades;
+ 0.000, - 0.025 for medium hard and harder grades.

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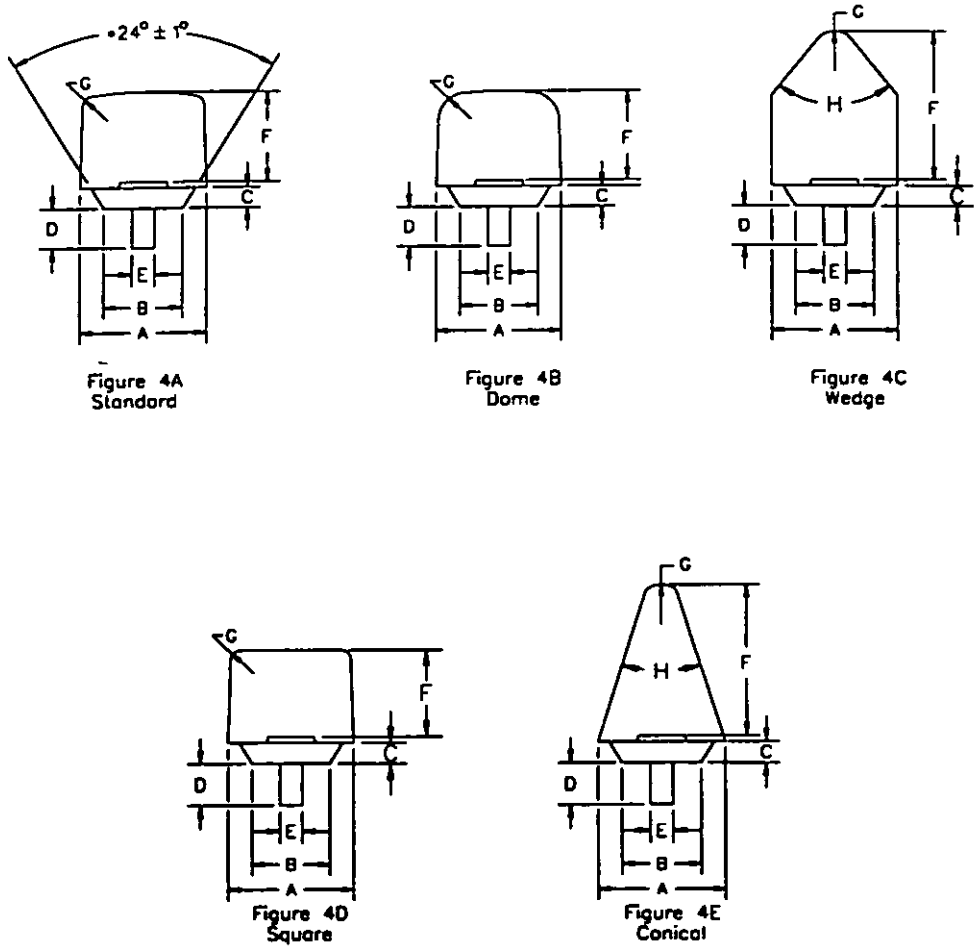


Figure 4A Standard

Figure 4B Dome

Figure 4C Wedge

Figure 4D Square

Figure 4E Conical

• THIS ANGLE IS TYPICAL FOR ALL SIZES AND SHAPES OF INSERTS.

Figure 4. Type I, class 3, style D plastic inserts.

3.12.3.1.3 Holder (head and handle). The head shall be of round cross-sectional shape, shall be of one-piece construction of aluminum alloy or zinc alloy, nonsparking and non-magnetic, except that the 8 and 10-3/4 pound hammers may be of ferrous metal. Each end shall have a tapered recess and shall be drilled and tapped to receive the plastic insert. The design shall be such that will ensure tightness of the inserts under all working conditions but shall allow the inserts to be readily removed, replaced, and interchanged. The head shall be fitted with a hickory handle conforming to 3.10.1 or fiberglass handle conforming to 3.9.3. The handles shall be secured with wedges conforming to 3.11 except the metal wedges shall be of a nonmagnetic materials, such as zinc alloy, when specified (see 6.2).

3.12.3.1.3.1 Dimensions of head of holder (without eye). The dimensions of the head identified by the letters A through G shall conform to Table XII for the diameter specified and shall be similar to Figure 5 (see 6.2).

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TABLE XII. Dimensions of type I, class 3, style D heads

A Head diameter ± 0.020 (Inches)	B Largest diameter of seat + 0.015 - 0.000 (Inches)	C Seat depth + 0.000 - 0.015 (Inch)	D Included angle ± 1 (Degrees)	E Thread diameter Nominal (Inch)	F Depth of full threads + 0.125 - 0.000 (Inch)	G Depth of hole (approx.) (Inches)
1	0.875	0.187	30	1/4	7/16	5/8
1-1/2	1.250	0.250	30	5/16	7/16	11/16
2	1.750	0.312	30	3/8	7/16	3/4
2-1/2	2.000	0.375	30	7/16	5/8	15/16
3	2.500	0.500	30	1/2	5/8	1-1/16

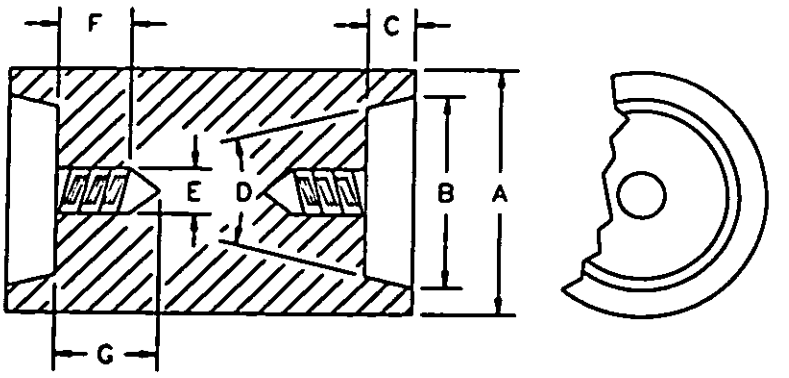


Figure 5. Identification of dimensions for type I, class 3, style D head (without eye).

3.12.3.1.4 Design a, regular. Type I, class 3, style D, design a, hammers shall conform to Table XIII for the weight, insert diameter, and length of head including standard faces specified and shall be similar to Figure 6 (see 6.2).

TABLE XIII. Type I, class 3, style D, design a hammers, inserted face, screw-in head, plastic inserts, regular

Weight 1/ (tolerances shown in ounces)	Diameter insert faces ± 1/16 heads ± 0.020 (Inches)	Length of head including standard inserts ± 3/16 (Inches)	Length of head without inserts + 0.060 - 0.030 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
6 ounces ± 1	1	3-3/8	2	11	2 or 3
9 ounces ± 1	1	3-3/8	2	11	2 or 3
3/4 pound ± 1-1/2	1-1/2	4-1/4	2-3/8	13	2 or 3
1-1/4 pounds ± 2	1-1/2	6-1/4	4-1/2	13	2 or 3
1-1/4 pounds ± 2	1-1/2	4-1/4	2-3/8	13	2 or 3

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TABLE XIII. Type I, class 3, style D, design a hammers, inserted face, screw-in head, plastic inserts, regular (cont'd)

Weight 1/ (tolerances shown in ounces)	Diameter insert faces ± 1/16 heads ± 0.020 (Inches)	Length of head including standard inserts ± 3/16 (Inches)	Length of head without inserts + 0.060 - 0.030 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
1-1/4 pounds ± 2	2	4-3/4	2-1/2	13	2 or 3
2 pounds + 3, - 2	2	4-3/4	2-1/2	13	2 or 3
3-1/4 pounds + 5, - 3	2	6-1/2	3-7/8	15	2 or 3
2-1/2 pounds + 4, - 3	2-1/2	6	3-1/4	15	2 or 3
4 pounds + 6, - 4	2-1/2	6	3-1/4	13	2 or 3
3-1/2 pounds + 8, - 5	3	6-3/4	3-3/4	14	2 or 3
6-1/2 pounds + 8, - 5	3	6-3/4	3-3/4	14	2 or 3
8 pounds + 10, - 5	3	7-3/8	4-5/8	28	2 or 3
10-3/4 pounds + 12, - 6	3	8-3/4	5-3/4	36	2 or 3

1/ Weight includes head, inserts, and handle.



Figure 6. Type I, class 3, style D, designs a and b hammers, inserted face, screw-in head, regular and no bounce, plastic inserts.

3.12.3.1.5 Design B, no bounce. Design b hammers shall conform to 3.12.3 through 3.12.3.1.3.1 except the head shall be designed to hold small metal pellets to restrict rebound during normal use. The head shall be of such design that the pellets will be retained within the head when the handle is removed or being inserted. Design b hammers shall conform to Table XIV for the weight specified and shall be similar to Figure 6 (see 6.2).

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TABLE XIV. Type I, class 3, style D, design b hammers, inserted face, screw-in head, plastic inserts, no bounce

Weight 1/ (tolerances shown in ounces)	Diameter insert faces ± 1/16 heads ± 0.020 (Inches)	Length of head including standard inserts ± 3/16 (Inches)	Length of head without inserts + 0.060 - 0.030 (Inches)	Handle length ± 1 (Inches)	Eye (Number)
1 pound ± 2	1-1/2	5-1/4	3-1/2	13	2 or 3
1-1/2 pounds + 3, - 2	1-1/2	6-1/4	4-3/8	13	2 or 3
2-1/2 pounds + 4, - 3	2	6-7/8	4-5/8	13	2 or 3
3-1/2 pounds + 5, - 3	2	8-3/4	6-1/2	14	2 or 3
5-1/2 pounds + 7, - 4	3	8	5	14	2 or 3

1/ Weight includes head, inserts, and handle.

3.13 Type II hammers, solid head.

3.13.1 Class 1, copper. The head of the copper hammers shall be square or barrel shaped and shall be of not less than 99 percent copper. Square heads shall have chamfered corners on the upper and lower portions. Both striking faces shall be flat and perpendicular to the longitudinal axis of the head and shall have a surface finish roughness height value less than 200 microinches when calculated as an arithmetic average. Other areas shall have a surface Finish conforming to good commercial practice. The hammers shall be inspected prior to lacquering or coating. The head shall be fitted with a hickory handle conforming to 3.10.1 or fiberglass handle conforming to 3.10.3, or when specified (see 6.2), steel handles conforming to 3.10.2 shall be furnished. Wood handles shall be secured with wedges conforming to 3.11. The shape of the head and the eye number shall be optional with the contractor unless otherwise specified in the contract or order (see 6.2). Eye numbers do not apply to heads with steel handles.

3.13.1.1 Type II, class 1 copper hammers. Type II, class 1 copper hammers shall conform to Table XV for the weight specified and shall be similar to Figure 7a or 7b (see 6.2).

TABLE XV. Type II, class 1 copper hammers

Weight 1/ (tolerances shown in ounces) (Pounds)	Length of head (Inches)	Diameter of head or distance across flats (Inches)	Handle length (overall) (Inches)	Eye		
					Dimensions (minimum)	
				Number	B (Inch)	C (Inches)
1/2 ± 2	2-1/4 ± 3/4	7/8 + 3/8, - 1/8	11 ± 1	2 or 3	5/16	9/16
1 ± 2	3 ± 3/4	1 + 3/8, - 1/8	12 ± 1	2 or 3	5/16	9/16
1-1/2 ± 3	3 ± 3/4	1-1/4 + 3/8, - 1/8	12 ± 1	2 or 3	5/8	7/8
2 ± 3	3-1/4 ± 3/4	1-1/4 + 3/8, - 1/8	13 ± 1	2 or 3	5/8	7/8
3 ± 5	4 ± 3/4	1-1/2 + 3/8, - 1/8	14 ± 1	2 or 3	3/4	1
4 ± 5	4-1/2 ± 3/4	1-3/8 + 3/8, - 1/8	15 + 2, - 1	2 or 3	3/4	1
6 ± 5	5 ± 3/4	2 + 3/8, - 1/8	18 + 2, - 1	2 or 3	3/4	1

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TABLE XV. Type II, class 1 copper hammers (cont'd)

Weight 1/ (tolerances shown in ounces) (Pounds)	Length of head (Inches)	Diameter of head or distance across flats (Inches)	Handle length (overall) (Inches)	Eye		
					Dimensions (minimum)	
				Number	B (Inch)	C (Inches)
8 ± 8	6 ± 1	2 + 3/8, - 1/8	19 + 2, - 1	2 or 3	1	1-1/4
10 ± 8	7 ± 2	2 + 3/8, - 1/8	20 + 2, - 1	2 or 3	1	1-1/4
12 ± 8	8 ± 2	2 + 7/8, -1/8	22 + 2, - 1	2 or 3	1	1-1/4

1/ Weight includes head and handle.

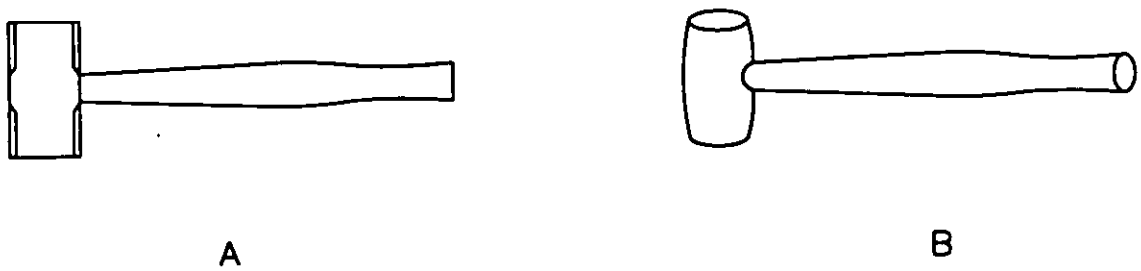


Figure 7. Type II, class 1 copper hammer.

3.13.2 Class 2, lead. The head of lead hammers shall be antimonial lead having a Brinell hardness of 10 to 16. The head shall be barrel shaped and shall have a smooth natural finish all over. The head shall have a steel handle conforming to 3.10.2

3.13.2.1 Type II, class 2 lead hammers. Type II, class 2 lead hammers shall conform to Table XVI for the weight specified and shall be similar to Figure 8 (see 6.2).

TABLE XVI. Type II, class 2 lead hammers

Weight 1/ (tolerances shown in ounces) (Pounds)	Diameter of head ± 1/8 (Inches)	Handle length (overall) ± 1 (Inches)
1 + 3, - 1	1	12
1-1/2 + 3, - 2	1-1/4	12
2-1/2 + 4, - 2	1-1/4	13
3 + 4, - 3	1-1/2	14
3-1/2 + 4, -3	1-1/2	14
5 + 6, - 4	1-3/4	16
6 + 8, - 4	2	18

1/ Weigh includes head and handle.

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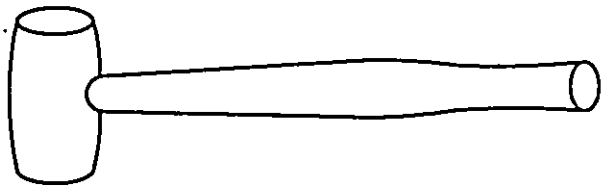


Figure 8. Type II, class 2 lead hammer.

3.14 Type III mallet.

3.14.1 Class 1, rawhide. The head of class 1 mallets shall be made of one piece of rawhide with not more than two filler strips. The hide shall be wound in a smooth, spiral, cylindrical shape. Joints at the end of the spiral shall be smooth and may be formed by a skived and glued lap, or a molded and pinned lap. The pin shall be not more than 3/16 inch off the center of the head. The striking face shall show no voids, shall be parallel within 1/32 inch and shall be 90 ± 2° to the sides of the head. The edge of the face shall be radiused or chamfered. The sides shall be smooth and free from air pockets, except that cracks or wrinkles not over 1/16 inch deep and at least 1/4 inch from the striking face will be permitted. The head shall be completely covered with a moisture-resistant coating. The head shall be fitted with a hickory handle conforming to 3.10.1 or fiberglass handle conforming to 3.10.3. The head may have a straight hole and be pinned or otherwise permanently fastened to the handle in lieu of taper hole and wedges.

3.14.1.1 Type III, class 1, rawhide mallets. Type III, class 1 rawhide mallets shall conform to Table XVII for the weight specified and shall be similar to Figure 9 (see 6.2).

TABLE XVII. Type III, class 1 rawhide mallets

Weight 1/ (Approx.) (Ounces)	Length of head + 3/8, - 1/4 (Inches)	Diameter of head ± 1/8 (Inches)	Handle length overall + 3, - 1 (Inches)
2	2-1/4	1	9
4	2-3/4	1-1/4	10
6	3	1-1/2	10
8	3-1/4	1-3/4	11
11	3-1/2	2	12
22	4-1/2	2-3/4	12
24	4-3/8	2-3/4	13

1/ Weight includes head and handle.

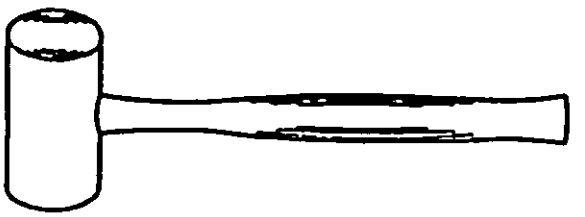


Figure 9. Type III, class 1 rawhide mallet.

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3.14.2 Class 2, plastic. The head of the class 2 mallets shall be composed of molded plastic and, except when otherwise specified herein, the head shall withstand the tests specified in 4.4.2 for plastic insert faces. The plastic heads shall be of four individual hardness grades with each grade having an identifying color. The heads shall conform to Table XVIII for the hardness name specified (see 6.2). The head shall be fitted with a hickory handle conforming to 3.10.1 or fiberglass handle conforming to 3.10.3. The head may have a straight hole and shall be pinned or otherwise permanently fastened to the handle in lieu of a taper hole and wedges. Both faces shall be straight or slightly convex and the sides shall be straight or slightly tapered toward the faces. The head shall have a smooth natural finish.

TABLE XVIII. Type III, class 2, plastic mallet head hardness grades

Hardness name	Hardness color	Test instrument and reading	Hardness range
Soft	Brown	Shore A, 30 second	45 to 58
Medium	Red	Shore A, 30 second	59 to 72
Tough	Green	Shore A, 30 second	73 to 85
Hard 1/	Black	Shore D, 30 second	49 to 65

1/ Test of 4.4.2.9, permanent distortion, not applicable.

3.14.2.1 Type III, class 2 plastic mallets. Type III, class 2 plastic mallets conform to Table XIXX of the weight specified and shall be similar to Figure 10 (see 6.2).

TABLE XIX. Type II, class 2 plastic mallets

Weight 1/ + 15 percent - 10 percent (Ounces)	Length of head + 3/8 - 1/4 (Inches)	Diameter of head + 1/8 - 1/16 (Inches)	Handle length (overall) ± 1 (Inches)
6	3-1/4	1-1/2	11
8	3-1/4	1-3/4	12
12	3-1/2	2	12
24	4-1/4	2-5/8	13

1/ Weight includes head and handle.

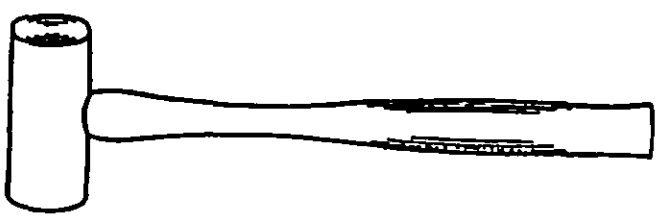


Figure 10. Type III, class 2, plastic mallet.

3.14.3 Class 3, rubber. The head of class 3 mallets shall be barrel shaped, of solid resilient rubber, suitable for automotive body, fender, and battery work where marring of the work is to be avoided. Both faces shall be either flat or slightly convex and the head shall have a smooth, natural finish. The head shall have a hickory handle conforming to 3.10.1. The head shall be securely and permanently bonded with the handle in such a manner that

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there is no separation of rubber from the handle under the most severe working conditions. The rubber head shall conform to the physical requirements of 3.8 and Table II, and shall withstand the test specified in 4.419.

3.14.3.1 Type III, class 3 rubber mallets. Type III, class 3 rubber mallets shall conform to Table XX and shall be similar to Figure 11.

TABLE XX. Type III, class 3 rubber mallet

Weight 1/ ± 5 (Ounces)	Length of head ± 5/8 (Inches)	Diameter of head + 7/8 - 1/4 (Inches)	Handle length overall + 1 - 2 (Inches)
24	4	2	15

1/ Weight includes head and handle.

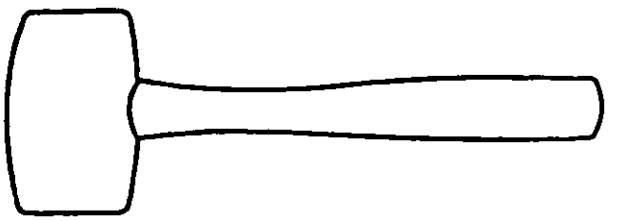


Figure 11. Type III, class 3 rubber mallet.

3.15 Type IV, adapter, screw-in head. Adapters shall be of a high grade steel and fully heat treated to be suitable for use with portable power tools. One end shall have a tapered recess and shall be drilled and tapped to receive either a 1-1/2 or 2-inch nominal diameter plastic insert. The design shall be such that will ensure tightness of the insert under all working conditions but shall allow the insert to be readily removed, replaced, and interchanged. The adapters shall have a ground or machined finish and shall be protected against corrosion by zinc plating.

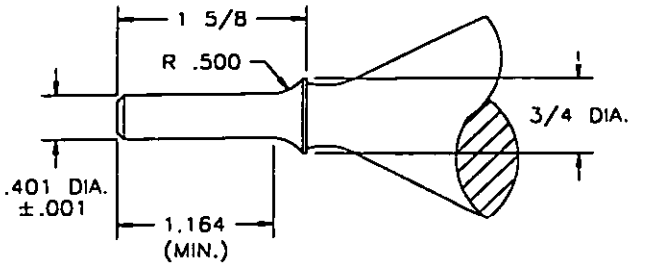
3.15.1 Plastic inserts for adapters. The 1-1/2 and 2 inch nominal diameter plastic inserts shall conform to 3.12.3.1 through 3.12.3.1.2.

3.15.2 Dimensions of adapter. The dimensions of the adapter identified by the letters A through F shall conform to Table XXI for the dimensions specified and shall be similar to Figure 12 (see 6.2). Dimensions without tolerances shown on the figure are approximate.

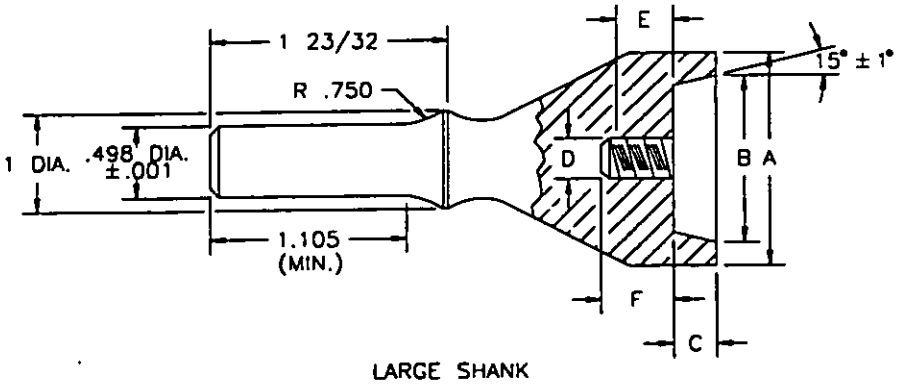
TABLE XXI. Dimensions of type IV adapters, screw-in head

Shank size ± 0.001 (Inch)	A Head diameter ± 0.020 (Inches)	B Largest diameter of seat + 0.000 - 0.015 (Inches)	C Seat depth + 0.000 - 0.015 (Inch)	D Thread diameter NC-2 Nominal (Inch)	E Depth of full threads + 0.125 - 0.000 (Inch)	F Depth of hole (approx.) (Inch)
0.401	1-1/2	1.250	0.250	5/16	7/16	11/16
0.401	2	1.750	0.312	3/8	7/16	3/4
0.498	1-1/2	1.250	0.250	5/16	7/16	11/16
0.498	2	1.750	0.812	3/8	7/16	3/4

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SMALL SHANK



LARGE SHANK

Figure 12. Type IV adapter, screw-in head.

3.16 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.17 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall bc free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.2.1 Inspection of the end item. Inspection of the hammers shall be performed as specified in the following paragraphs. A lot shall consist of hammers, mallets, adapters, or inserts of the same type, class, and style. The sample unit shall be one completely fabricated and assembled hammer, mallet, adapter, or insert.

4.2.1.1 Visual examination. Each sample unit shall be visually examined for the defects set forth in Table XXII. The inspection level shall be level S-2, with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE XXII. Classification of defects

Examine	Defect
Finish	Not as specified Ground surfaces, rough, deep scratches, unevenly finished Coating not smooth, continuous, adherent, blisters, runs, sags, or other detrimental defects to protective value Cast, molded, extruded, rolled, or forged surfaces not free of scale, rust, or other harmful defects Rubber surfaces, not smooth, pock marks, blisters, folded over lamination, or foreign material embedded in surface
Material	Not fabricated from specified materials
Design	Does not conform to type, class, or style specified
Construction and Workmanship	Hammer not complete, parts missing, improperly fitted or assembled, damaged or nonconforming Inserts not readily removed, replaced and interchangeable Evidence of porosity, cracks, burrs, fins, or sharp edges Threaded studs not securely attached to base of insert, not coated for corrosion resistance, or evidence of non-adherence Studs threads damaged or incomplete Threads in head damaged or incomplete Wood handles not straight grain, not smoothly finished, splits, cracks, or worm holes Handle not secured and fitted to head as required, wedge or pin missing Metal handles not knurled Wedges not flush with front surface of tool eye End of spiral on rawhide mallets not smooth or securely joined Voids in face of rawhide mallet Operation omitted or improperly performed
Marking	Missing, incorrect or illegible Plastic inserts not color codes as required

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4.2.1.2 Dimensional examination. Examination shall be made for dimensional requirements specified in applicable paragraphs and tables. Any nonconformance shall constitute a defect. The inspection level shall be S-2 with an AQL of 2.5 defects per hundred units.

4.2.1.3 Testing of rubber. When specified (see 6.2), the manufacturer shall provide a sheet of rubber material 0.07 to 0.08 inch thick having a total surface area of at least one square foot, together with at least eight solid heads of insert face blanks which shall be of the same composition and cure, and prepared at the same time from the material used in the heads of inserts furnished. Each sample rubber insert shall be tested as specified in 3.7 and Table I. Each sample rubber mallet shall be tested as specified in 4.4.1.9 through 4.4.1.9.6 to verify compliance with the physical properties specified in 3.7.1 and Table II.

4.2.1.4 Testing of plastic. When specified (see 6.2), the manufacturer shall provide at least eight solid heads or insert face blanks (specimens) of the plastic material which shall be of the same composition and cure and prepared at the same time as the material used in the heads or insert furnished. Each sample shall be tested as specified under 4.4.2 to verify compliance with this specification.

4.2.1.5 Pull-apart, striking, and temperature test. Testing shall be performed on each sample hammer (when applicable) for determination of requirements as specified in 4.4.2.6 and 4.4.2.12. The hammers shall be kept at the temperature specified in 3.9.5 for at least 5 hours and be allowed to return to a temperature of $70^{\circ} \pm 10^{\circ}$ prior to conducting tests. The inspection level shall be S-2 with and AQL of 2.5 defects per hundred units.

4.3 Preparation for delivery. Inspection to determine compliance with requirements for preparation of delivery shall be performed in accordance with PPP-P-40.

4.4 Test procedures.

4.4.1 Tests for rubber inserts. Rubber for inserts shall be tested in accordance with 4.4.1.1 through 4.4.1.7.

4.4.1.1 Tensile strength and elongation. Tensile strength and ultimate elongation shall be determined on specimens by the procedure described in FED-STD-601.

4.4.1.2 Hardness. Hardness measurements shall be made on the mallet head or insert face with the Shore A instrument. The average of measurements on duplicate specimens shall be reported.

4.4.1.3 Resilience test. Test specimens shall be discs 1/2 inch thick by at least 1-1/4 inch diameter and shall be buffed flat on both sides. Insert face blanks of the same width may be used.

4.4.1.3.1 Apparatus. The sample shall be positioned so that the point of impact on the plunger shall be at the point of maximum thickness of the section of the sample to be tested. The sample shall be held in position by the weight of the adjustable unit of resiliometer. The specimen shall be given a warmup to 20 impacts at approximately 2 impacts per three seconds. The average of the next 10 readings shall be taken as the resilience of the specimen. Testing shall be done at $82^{\circ}F$ ($28^{\circ}C$) $\pm 5^{\circ}F$ after the sample shall have been conditioned for at least 30 minutes at this temperature.

4.4.1.4 Determination of tear resistance.

4.4.1.4.1 Test specimen. The test specimens shall be cut from sheets approximately 0.08 inch thick by 6 inches square by means of a Graves angle die.

4.4.1.4.2 Apparatus. The apparatus used for the termination of tear resistance shall be a standard Scott rubber tester.

4.4.1.4.3 Procedure. The specimen shall be clamped in the jaws of the testing machine so that the longitudinal axis is in line with the direction of the application of the load. The load shall be applied uniformly. Resistance shall be expressed in pounds pull per inch thickness.

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4.4.1.5 Swelling test.

4.4.1.5.1 Test specimens. The test specimens shall be rectangular synthetic rubber blocks 2 by 1 inch and not over 1/16 inch thick. Three specimens shall be used for each test and the results averaged.

4.4.1.5.2 Procedure. The volume of each test specimen shall be measured by the water displacement method, in which the specimen is accurately weighed to the nearest mg in air (W1) and in distilled water (W2) at room temperature. When weighing in water, care shall be taken that the specimen is free from adhering air bubbles, and, if necessary, it may first be wetted by dipping into to 95 percent ethyl alcohol and thoroughly rinsed with distilled water. After weighing, the specimen shall be blotted dry with filter paper, completely immersed in ASTM Oil No. 3 in a flask, and allowed to stand for 46 ± 1/4 hours at 158°F. Excessive evaporation shall be avoided by fitting the container with a cork stopper. At the termination of the immersion period, the specimen shall be removed from the special testing fuel, dipped in 95 percent alcohol, blotted lightly with filter paper, and placed in a tared weighing bottle and weighed (W3). The specimen shall then be removed from the bottle and weighed in distilled water (W4) in immediate consecutive procedure to determine the displacement after the test. The final weighing shall be completed within 3 minutes after removal from the special testing fuel.

4.4.1.5.2.1 Calculation. The increase in volume shall be calculated as follows:

$$\text{Percentage increase in volume} = \frac{(W3 - W4) - (W1 - W2) \times 100}{W1 - W2}$$

4.4.1.6 Cut resistance. Test specimens shall be 1/2 x 1 x 1 inch which have been dusted lightly with talc. Insert-face blanks may be used after being cut to this size.

4.4.1.6.1 Apparatus. The cutter shall be the same as that normally supplied with the Bell Telephone compression cutter tester, the cutter shall be wedge shaped having an included angle of 22°. At the apex of the wedge, a steel music wire 0.046 inch in diameter shall be imbedded.

4.4.1.6.2 Procedure. The specimen shall be placed on the anvil and the cutter brought down against the specimen at a rate of 2 inches per minute. Cutting will be evidenced by a sharp decrease in the indicated load.

4.4.1.7 Impact resistance test. Impact resistance is defined in this case as the resistance to permanent indentation by a free falling weight weighing 10 pounds with a hemispherical indentor 1/2 inch in diameter on its striking surface, the weight falling from a height of 5 feet. The weight shall be cylindrical in shape and shall be dropped through a piece of seamless tubing slightly larger than the diameter of the weight in order to ensure that the full force of the falling weight will be acting normal to the striking surface. Test specimens shall be 1/2 inch thick by at least 1-1/4 inch diameter and placed upon a heavy steel plate or similar foundation. Insert-face blanks may be used. The rubber material shall show no sign of chipping or cracking after the impact test.

4.4.1.8 Rubber insert or head. The rubber insert or head shall be free from patches, cracks, holes, blisters, decontaminations, soft spots, or any other defect not specifically permitted by the contract or order.

4.4.1.9 Test for rubber mallet heads. Rubber for rubber mallet heads shall be tested in accordance with 4.4.1.9 through 4.4.1.9.6 to verify compliance with 3.7.1 and Table II.

4.4.1.9.1 Tensile strength and elongation. Tensile strength and ultimate elongation shall be determined in accordance with 4.4.1.1.

4.4.1.9.2 Hardness. Hardness measurements shall be made in accordance with 4.4.1.2.

4.4.1.9.3 Resilience test. Resilience test shall be made in accordance with 4.4.1.3 and 4.4.1.3.1.

4.4.1.9.4 Tear resistance. Determination of tear resistance shall be made in accordance with 4.4.1.4 through 4.4.1.4.3 except that ASTM Tear Die "C" may be used in lieu of the Graves Angle Die specified in 4.4.1.4.1.

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4.4.1.9.5 Swelling. The swelling test shall be made in accordance with 4.4.1.5 through 4.4.1.5.2.1.

4.4.1.9.6 Rubber mallet heads. The rubber mallet head shall be in accordance with 4.4.1.8.

4.4.2 Test for plastic insert faces and mallet heads.

4.4.2.1 Insert faces and mallet heads. Plastic for insert faces and mallet heads shall be tested in accordance with the tests specified in 4.4.2.2 through 4.4.2.9. A new specimen shall be used for each test.

4.4.2.2 Flammability test. The sample shall be clamped in a support in a horizontal position. Under the sample there shall be clamped a piece of 10-mesh Bunsen burner gauze in a horizontal position, 1/4 inch below the bottom of the sample with approximately 1/2 inch of the sample extending beyond the edge of the gauze. A Bunsen burner or alcohol lamp, with a flame of 1/2 inch to 3/4 inch in height shall be placed under the free end of the sample and adjusted so that the flame tip is just in contact with the sample. At the end of 30 seconds the flame shall be removed and the plastic allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the sample. Should burning cease after the first ignition, the Bunsen burners shall be replaced under the free end immediately following the extinction of the flame for a second period of 30 seconds. If the sample does not continue to burn it shall be considered as self-extinguishing. If the sample continues to burn, the burning rate shall not exceed 2 inches per minute.

4.4.2.3 Heat distortion. The sample for test shall be suspended in boiling water at 212°F for a period of 2 minutes. At the end of this period it shall be removed from the water and shall show no significant distortion or blistering. Significant distortion or blistering (for purpose of this test) means an increase or decrease of at least 4 percent in the original dimensions of the sample, either of a uniform or irregular bulge, blister, or depression.

4.4.2.4 Impact test at room temperature. Samples shall be placed, screwed, or otherwise arranged to simulate actual use on a steel plate. A cylindrical weight for the specific size shall be dropped through a length of seamless tubing, slightly larger than the weight, to ensure full force of the weight acting on the normal striking face. The sample shall not break, crack, or significantly distort when the weight has been dropped a minimum of 20 times from a height determined by Table XXIII.

TABLE XXIII. Impact test at room temperature

Diameter of insert (Inches)	Weight (Pounds)	Drop (Inches)
1 to 1-1/4	10	28
1-1/2 to 1-3/4	10	33
2 to 2-1/4	15	38
2-1/2 to 2-5/8	15	45
3	25	42

4.4.2.5 Impact at -30°. A sample hammer inset or mallet head (except medium hard) shall be kept at a temperature of -30° ± 5°F for 5 hours and subjected to the tests described in 4.4.2.4, except that only one blow shall be struck. Weight and drop shall be in accordance with Table XXIV. The tests shall be completed within 1 minute after removal of the sample from the cooling medium or cabinet. Failure of more than 10 percent of the units tested will constitute rejection. A sample is deemed to have failed if it breaks or cracks.

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TABLE XXIV. Impact test at 30°

Face diameter (Inches)	Weight (Pounds)	Drop (Inches)
1	10	11
1-1/2 to 1-3/4	10	12
2	15	15
2-1/2 to 2-5/8	15	18
3	15	28

4.4.2.6 Striking test. Each sample hammer or mallet selected in accordance with 4.2.4 shall withstand 25 full swinging blows by continuous hand striking or the mechanical equivalent on the outside corner of heavy 2 by 2 angle iron or block of steel. The corner shall be smoothed to a radius of not less than 1/32 nor more than 3/64 inch. The angle iron or steel block shall be placed on an anvil, concrete or other solid base, to ensure maximum delivered blow to the tool being tested. The insert or head shall have failed if there are any permanent impressions, displacement of face material, flaking, mushrooming, or any splits or gouges over 1/32 inch deep, running more than 50 percent of the face diameter.

4.4.2.7 Solvent test. A test specimen shall be completely submerged in hydraulic oil in accordance with MIL-H-5606 and hydrocarbon fluid in accordance with TT-S-735 for a period of 24 hours in each solution. The specimen shall show no greater than 2 percent increase or 1 percent decrease in volume after removal from the solvent.

4.4.2.8 Hardness test (durometer). Samples shall be soaked for a minimum of 2 hours at 68° ± 2°F and the hardness tested within 1 minute after removal from cabinet. Using a durometer A-2 or D-2 instrument, as applicable, the specimen shall be placed on the instrument and pressure applied, the hardness reading shall be taken 30 seconds after contact, except that for the super soft grade an instantaneous reading shall be taken. Durometer hardness shall fall within tolerances shown in 3.11.1.4, 3.11.2.1, Table X or XVIII as applicable.

4.4.2.9 Permanent distortion test. Samples shall be soaked for 2 hours at 68° ± 2° F and tested within 2 minutes as follows: Samples shall be clamped in a vise against the threaded surface of a national course thread series 1/2 inch bolt. The maximum flat or radius at the crest of the threads shall not exceed 0.010 inch. The vise shall be tightened until the bolt imbeds itself 1/4 inch into the head or insert face and after being held in this position for a period of 1 minute, the vise shall be loosened and the specimen removed. After a period of not less than 30 seconds or longer than 10 minutes the specimen shall not evidence any permanent marring or distortion and will have returned to its original shape. Material returning to shape in less than 30 seconds of which requires more than 10 minutes to recover shall be considered unsatisfactory.

4.4.2.10 Samples. Samples shall be free from patches, cracks, holes or any other defects, not specifically permitted by the contractor or order that may effect their serviceability.

4.4.2.11 Plastic inserts. Plastic inserts shall be considered to have failed if after heat distortion, solvent, or striking tests, the insert cannot be properly attached to the body as originally installed.

4.4.2.12 Pull-apart test. Each hammer selected in accordance with 4.2.4 that has the handle assembled with a chemical adhesive bond shall have the head and handle securely held in a standard tensile-testing machine, or some other method may be employed. Hammers with head weighing over 12 ounces, nominal weight, shall not pull apart or loosen when subjected to a pulling pressure of 2,250 pounds. Hammers with heads weighing 12 ounces or under, nominal weight, shall not pull apart or loosen when subjected to a pulling pressure of 750 pounds.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Inserts for hammers and adapters, and solid head hammers and mallets are designed to prevent marring or damaging of work of many varieties. Since many factors are involved, it is considered impractical to designate specific tools for specific work. Therefore, the following is offered as a guide in selecting the most suitable tool for the work:

(a) Maximum life of the tool may be obtained by using the softest grade of insert tip or mallet that will accomplish the work.

(b) Watch for these signs:

(1) When tool marks the work, select the next grade softer.
(2) When the tool face wears excessively, select the next grade softer or select a similar weight tool in the next series larger diameter.
(3) When the tool face does not drive or accomplish the work, select the next grade harder.
(4) For hammers with replaceable insert, consider using two different hardness inserts as required.

(c) Proper tools may be selected for work requiring use of nonsparking or nonmagnetic tools.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, style, design, and weight, if applicable, of item required (see 1.2 and applicable tables).
c. Type of handle required; wood (state whether A or AA), fiberglass or steel, except for type II, class 2 (see 3.10).
d. For type I, class 3, style D, design a, specify insert diameter (see 3.12.3.1.2); specify length of head including standard inserts (see Table XIII); specify hardness name of plastic insert (see 3.12.3.1.1); specify when nonmagnetic studs are required (see 3.12.3.1); specify shape when other than standard shape insert is required (see 3.12.3.1.2); specify when nonmagnetic metal wedges for the handle are required (see 3.12.3.1.3).
e. For type II, class 1, specify when steel handle is required (see 3.13.1); specify when particular shape of head is required (see 3.13.1); specify when particular eye number is required (see 3.13.1).
f. For type III, class 2 specify hardness name of plastic head (see 3.14.2).
g. For type IV adapters, specify shank and head diameter (see 3.15.2).
h. When only inserts are required, specify type, class, style, and diameter; for type I, class 3, style D, and type IV inserts specify diameter, hardness name, and shape (if other than standard shape is required).
i. When only holders are required, specify the type, class, weight (of the hammer) and insert face diameter; for type I, class 3, style D holders, specify weight (of the hammer), insert face diameter, and length of head without inserts (see Table XIII).
j. If chemical adhesive bond must withstand -60°F (see 3.10.5).
k. If a sheet of rubber material is required for testing (see 4.2.2.3).
j. Level of preservation, packaging, and packing required (see 5.1).

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6.2.1 Replacement fiberglass handle kits. Procurement document should specify the following:

a. Type, class, weight, handle length and eye number that replacement handles are for.
b. Temperature requirements (see 3.10.5 and 6.2 option b).
c. Quantity of kits required.

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Types I and II:

Rail:
Hammers, hand, not otherwsie indexed by name
Carload minimum weight 30,000 pounds
Motor:
Hammer, hand, not otherwise indexed
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification

Types III and IV:

Rail:
Tools, not otherwise indexed by name
Carload minimum weight 30,000 pounds

Motor:
Tools, hand, not otherwise indexed
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification

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